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HONORS CHEMISTRY • UNIT 3

Chemical Bonding

Why atoms stick together, how they share or trade electrons, and how the shape and polarity of a molecule decide how a substance behaves.

HOW TO USE THIS PACKET

Write definitions in your own words, draw every structure yourself and use the empty space for questions. LAB CORNER boxes are things you would see or do at a lab bench. The shape chart in 3.4 is the one page to keep for the rest of the year.

What we will learn

3.1	Types of chemical bonds: ionic and covalent
3.2	Properties of ionic and covalent compounds
3.3	Lewis diagrams for ionic and covalent compounds
3.4	Lewis structures & VSEPR theory (shapes and angles)
3.5	Bond polarity & molecular polarity
★	Extra practice & unit review

By the end of this unit I can...

- Explain what a full octet is and how atoms reach one by losing, gaining or sharing electrons.
- Predict whether a compound is ionic or covalent and describe the bond using metals/non-metals and electronegativity.
- Connect bonding type to properties: melting point, state at room temperature and electrical conductivity.
- Use Lewis dot diagrams to model ionic compounds and to draw Lewis structures for covalent molecules and polyatomic ions.
- Compare single, double and triple bonds in terms of bond length and strength.
- Use VSEPR theory to predict molecular shapes and bond angles.
- Use electronegativity, bond dipoles and shape to decide whether a molecule is polar or non-polar.

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TOPIC
3.1

Types of Chemical Bonds

Octet rule • ionic bonds • covalent bonds

BIG IDEA

Atoms bond to reach a **full outer shell of 8 electrons** (an **octet**; 2 for hydrogen and helium), the arrangement of the noble gases. They get there in one of two ways: **transfer** electrons (ionic bond) or **share** electrons (covalent bond).

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Octet	*How many electrons? Which elements are the exception?*	
Ionic bond	*Between what kinds of elements? What do the electrons do?*	
Covalent bond	*Between what kinds of elements? What do the electrons do?*	
Cation / anion	*Which one is formed by losing electrons?*	
Molecule	*Is NaCl a molecule? Why or why not?*	
Bond order (single / double / triple)	*How many electrons are shared?*	

THINK OF IT LIKE… TRADING CARDS VS. SHARING A TOY

An **ionic bond** is like a trade where one kid hands over a card for good and the two are now stuck together by the deal (opposite charges). A **covalent bond** is two kids sharing one toy: neither owns it outright, and the toy keeps them together.

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What does it mean to have a full octet?

1. In your own words, what is a full octet and why do atoms "want" one?

Ionic bonds

2. Which type of element typically forms a cation, and does it give or take electrons? Which type forms an anion, and does it give or take electrons?

CUES / KEY WORDS	HOW A SODIUM ATOM AND A CHLORINE ATOM FORM AN IONIC BOND
Metal **Non-metal** **Why does the ion form?**	

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Predicting ion charges

Use the periodic table. For each element: count the valence electrons, decide whether the atom loses or takes electrons, and give the ion's charge.

Element	# of valence e^-	Lose or take e^-?	Cation or anion?	Ion charge
Sodium				
Fluorine				
Aluminum				
Boron				
Nitrogen				
Sulfur				

QUICK RULE

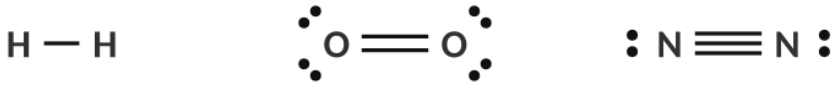
- **Group 1 → 1+, Group 2 → 2+, Group 13 → 3+.**
- **Group 15 → 3−, Group 16 → 2−, Group 17 → 1−.**
- Transition metals often form more than one charge (Fe^{2+} or Fe^{3+}); you will see these in Unit 4.

Covalent bonds

3. How is a covalent bond different from an ionic bond? Talk about the elements that form it and what happens to the electrons.

4. Some compounds contain single, double and triple bonds. What is the difference between them?

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single bond: H–H double bond: O=O triple bond: N≡N

Single, double and triple bonds.

Ionic or covalent?

SHORTCUT

metal + non-metal → **ionic**

non-metal + non-metal → **covalent**

(If you are unsure, compare electronegativity: a difference larger than about 1.7 means ionic. See 3.5.)

Compound	1st element: metal or non-metal?	2nd element: metal or non-metal?	Ionic or covalent?
NO_2			
$MgBr_2$			
CaO			
PI_3			
SO_2			
CaI_2			

More practice	Ionic or covalent?
KCl	
CCl_4	
Li_3N	
H_2O	
Al_2O_3	
CH_4	

More practice	Ionic or covalent?
NH_4Cl	

5. An ionic bond forms when…

A) two non-metals share electrons
B) a metal transfers electrons to a non-metal
C) two metals share electrons
D) two noble gases attract

6. How many electrons are shared in a double bond?

A) 2
B) 4
C) 6
D) 8

7. Which element is most likely to form a 2− ion?

A) Na
B) Cl
C) O
D) Mg

8. Which compound is covalent?

A) MgO
B) SO_2
C) KBr
D) $CaCl_2$

Name: _____ Date: _____ Period: _____

TOPIC
3.2

Properties of Ionic and Covalent Compounds

Melting point • state • conductivity • brittleness

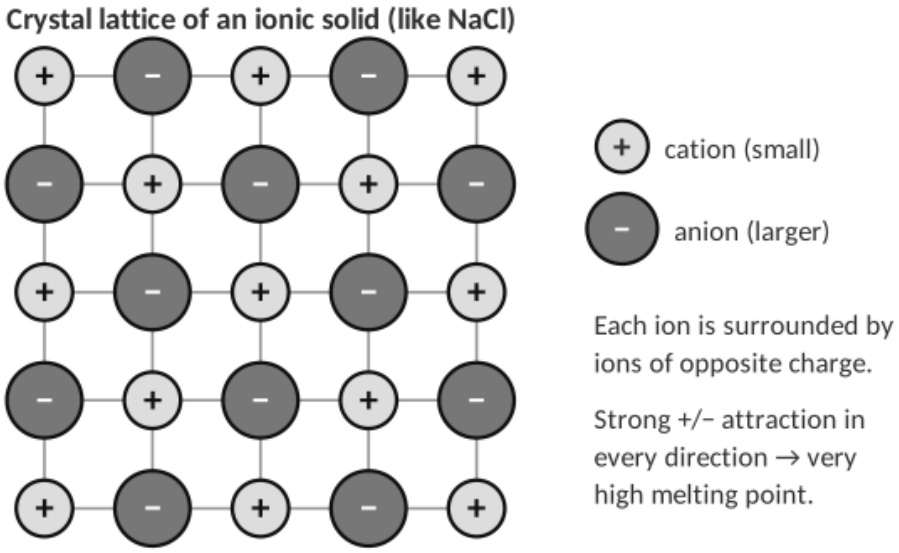
BIG IDEA

The kind of bond decides how a substance behaves: ionic compounds are hard, brittle, high-melting solids that conduct only when their ions can move; covalent (molecular) compounds usually melt at low temperatures and do not conduct.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Crystal lattice	*What holds it together?*	
Formula unit	*How is it different from a molecule?*	
Melting point	*What are you overcoming when a solid melts?*	
Electrical conductor	*What must be present?*	
Electrolyte	*Is a sucrose solution an electrolyte?*	

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Melting points tell you about the bond

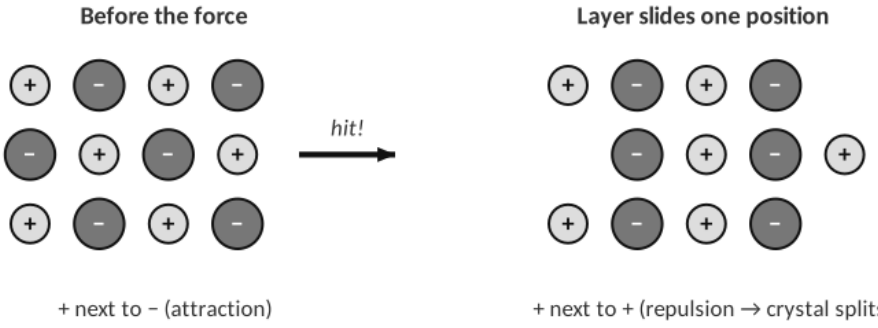
The table lists data for six substances. Classify each as ionic or covalent, then answer the questions that follow.

Substance	Ionic or covalent?	Melting point	State at room temp.
$CaCl_2$		772 °C	solid
H_2O		0 °C	liquid
MgO		2852 °C	solid
NH_3		−77 °C	gas
SrO		2531 °C	solid
H_2S			
$C_6H_{12}O_6$		146 °C	solid

1. Some compounds have a large difference in melting point. Use the type of bond to explain why the ionic compounds melt at much higher temperatures.

2. Based on this trend, do you expect H_2S to have a high or low melting point? Explain using the bond found in H_2S.

Why ionic crystals are brittle

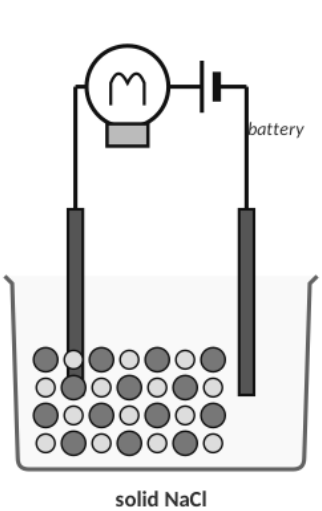


A hammer blow slides one layer of ions.

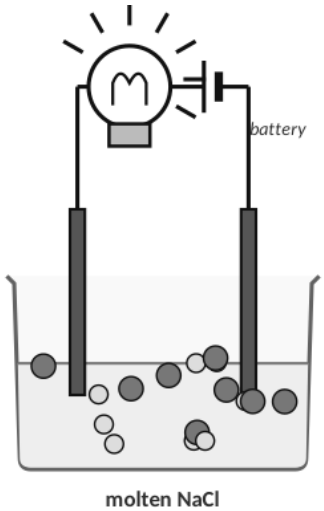
3. Explain why a salt crystal shatters when you hit it, while a metal bends.

LAB CORNER • Conductivity testing

A student uses a conductivity tester (two electrodes, a battery and a light bulb) to test substances. If current can flow through the sample, the bulb lights.

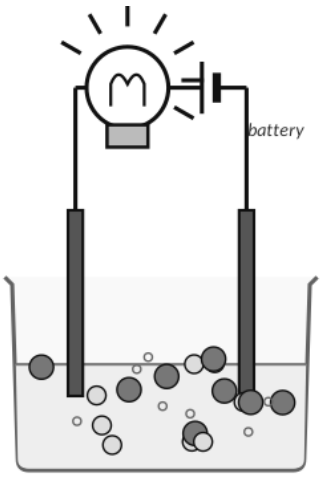


A: solid NaCl



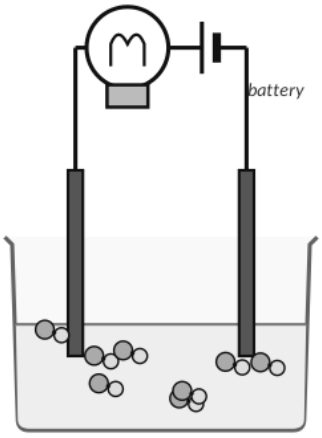
B: molten NaCl

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NaCl dissolved in water

C: NaCl dissolved in water



sugar dissolved in water

D: sucrose dissolved in water

[IMAGE GOES HERE]

Optional: conductivity beakers from the homework

4. For a substance to conduct electricity, what must be present?

5. Which beakers should the student expect to light the bulb? Explain.

6. For the beakers you did not choose, explain why they do not conduct.

7. Predict: would distilled water light the bulb? Would tap water? Explain.

Property	Ionic compounds	Covalent (molecular) compounds
Particles		
Melting / boiling point		
State at room temp.		
Conducts as a solid?		
Conducts when molten or dissolved?		
Hardness		

8. Which substance would conduct electricity when melted but not as a solid?

A) $C_6H_{12}O_6$
B) MgO
C) H_2O
D) NH_3

9. Why does $CaCl_2$ have a much higher melting point than H_2O?

A) covalent bonds are stronger
B) ionic attractions among many ions are strong
C) water has no bonds
D) $CaCl_2$ is a gas

10. Which of these solutions does NOT conduct electricity?

A) KBr(aq)
B) NaOH(aq)
C) $C_{12}H_{22}O_{11}$(aq)
D) HCl(aq)

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TOPIC
3.3

Lewis Diagrams for Ionic and Covalent Compounds

Electron transfer • shared pairs • bond length

BIG IDEA

A **Lewis diagram** is a bookkeeping tool for valence electrons. For ionic compounds it shows electrons **moving** from metal to non-metal; for covalent compounds it shows **shared pairs** (bonds) and **lone pairs**.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Lewis dot diagram (atom)	*How many dots does carbon get?*	
Bonding pair	*How is it drawn?*	
Lone pair	*Which atom owns it?*	
Formula unit (ionic)	*How do you find the ratio of ions?*	
Bond length	*What is the trend for single, double and triple?*	

Ionic compounds: show the electron transfer

STEPS

1. Draw each atom with its valence electrons as dots.
2. Draw an arrow from each metal to the non-metal for every electron transferred.
3. Draw the ions in brackets: the cation shows NO dots (its outer shell is gone); the anion shows 8 dots.
4. Write the charge outside the bracket, and use as many metal atoms as it takes so that the electrons given = electrons taken.

WORKED EXAMPLE — POTASSIUM OXIDE

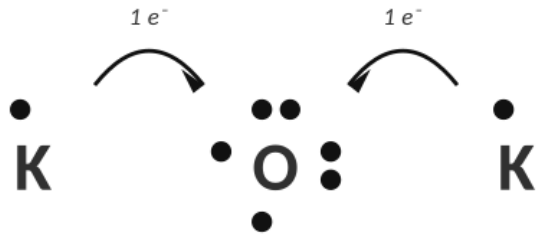
Use Lewis dot diagrams to show why potassium oxide is K_2O.

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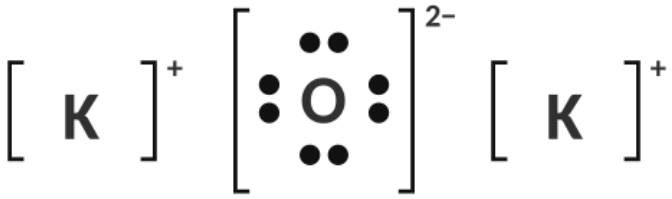
Work it out here:

Answer: _____

BEFORE: neutral atoms



AFTER: ions (full octets)



Potassium oxide: two K^+ for each O^{2-}.

Your turn

Draw Lewis dot diagrams (with arrows and brackets) for each compound and write its formula.

1. Calcium oxide: calcium + oxygen

Diagram and formula:

2. Lithium nitride: lithium + nitrogen. (The label on a bottle has worn off, but the student knows it contains lithium and nitrogen.)

Diagram and formula:

3. Magnesium chloride: magnesium + chlorine.

Diagram and formula:

4. Both K_2O and CaO contain oxygen. Explain why one formula has a subscript 2 and the other doesn't.

Covalent compounds: Lewis structures step by step

STEPS

1. **Count total valence electrons** (add one electron for each negative charge; subtract one for each positive charge).
2. **Skeleton**: put the least electronegative atom in the middle (never H, never F). Connect the atoms with single bonds (2 e^- each).
3. **Outer atoms first**: add lone pairs to the outside atoms until each has 8 (H gets 2).
4. **Leftover electrons** go on the central atom.
5. If the central atom has fewer than 8, **move a lone pair from an outer atom** to make a double (or triple) bond.
6. Check: every atom has an octet (H has 2), and the electron count matches step 1.

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WORKED EXAMPLE — WATER

Draw the Lewis structure of H_2O.

Work it out here:

Answer: _____

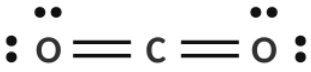


WORKED EXAMPLE — CARBON DIOXIDE

Draw the Lewis structure of CO_2.

Work it out here:

Answer: _____



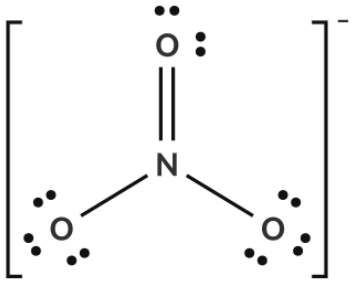
WORKED EXAMPLE — NITRATE ION

Draw the Lewis structure of NO_3^-.

Work it out here:

Answer: _____

Name: _____ Date: _____ Period: _____



Count the electrons

Species	Valence-electron count	Total
NH_3		
CH_4		
HCN		
SO_2		
PCl_3		
NH_4^+		
CO_3^{2-}		

Practice: draw the Lewis structures

Molecule	Lewis structure (draw it here)
NH_3	
CH_4	

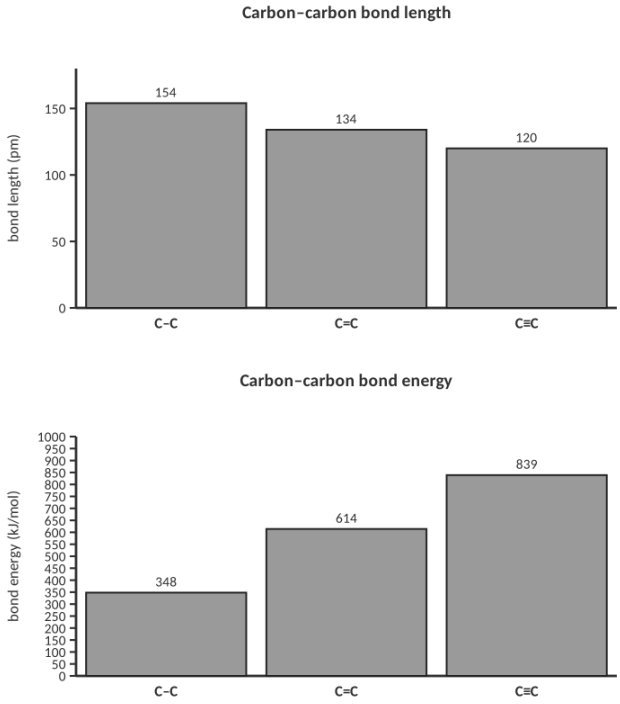
Molecule	Lewis structure (draw it here)
HCN	
SO_2	
CH_2O	
C_2H_4	

Bond length and bond strength

A student is interested in the bond lengths of ethane (C_2H_6), ethene (C_2H_4) and ethyne (C_2H_2). Each carbon must have 8 electrons (4 bonds in total) and each hydrogen 2.

Compound	C–C bond type	Lewis structure	Bond length (pm)	Bond energy (kJ/mol)
C_2H_6			154	348
C_2H_4			134	614
C_2H_2			120	839

5. Which of the carbon–carbon bonds is the shortest? Explain using the attraction between protons and electrons.



6. Describe the relationship between the number of shared pairs, bond length and bond energy.

7. Which is the correct total number of valence electrons for the Lewis structure of PO_4^{3-}?

A) 29
B) 32
C) 30
D) 26

8. How many lone pairs are on the central atom of H_2O?

A) 0
B) 1
C) 2
D) 3

9. A compound is made of Mg^{2+} and N^{3-}. What is the formula?

A) MgN
B) Mg_2N_3
C) Mg_3N_2
D) Mg_3N

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TOPIC
3.4

Lewis Structures & VSEPR Theory

Electron domains • shapes • bond angles

BIG IDEA

VSEPR (Valence Shell Electron Pair Repulsion) says that electron pairs around a central atom repel each other, so they spread out as far apart as possible. The resulting arrangement of the **atoms** is the shape of the molecule.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Electron domain	*What counts as one domain?*	
Electron geometry	*Do lone pairs count?*	
Molecular shape	*Do lone pairs count?*	
Bond angle	*How is it measured?*	
VSEPR theory	*Why do molecules have shapes?*	

THINK OF IT LIKE… TIED-TOGETHER BALLOONS

Tie 2 balloons together and they point in opposite directions (180°). Tie 3 and they make a flat triangle (120°). Four balloons form a tetrahedron (109.5°). Balloons are electron domains: they push each other apart as far as the knot allows. A fatter balloon (a lone pair, which takes more room than a bond) squeezes the others slightly closer.

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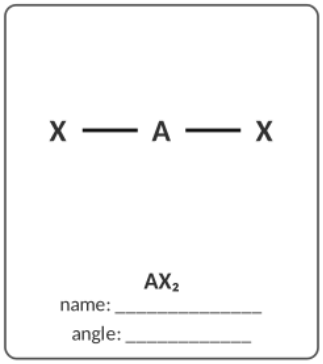
In your own words: what is the big idea of VSEPR?

HOW TO NAME A SHAPE

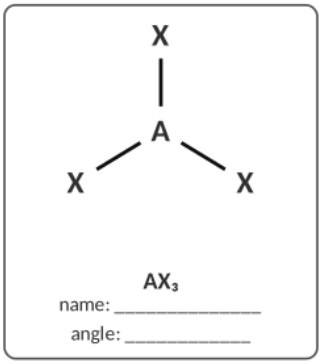
1. Draw the Lewis structure.
2. Count the electron domains on the **central atom** (each bond, whether single, double or triple, counts once; each lone pair counts once).
3. Count how many of the domains are lone pairs (E).
4. Use the chart below: the number of domains gives the electron geometry; the lone pairs remove positions to give the molecular shape.

Name: _____ Date: _____ Period: _____

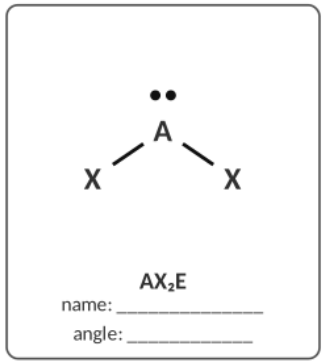
The VSEPR chart (fill in the name and angle of every shape)



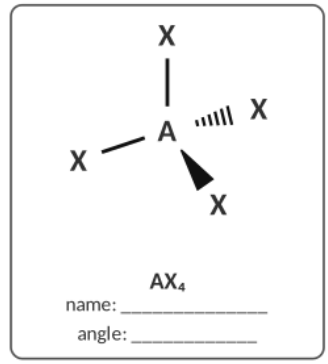
AX_2
name: _____
angle: _____



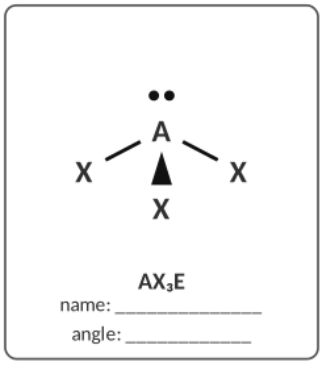
AX_3
name: _____
angle: _____



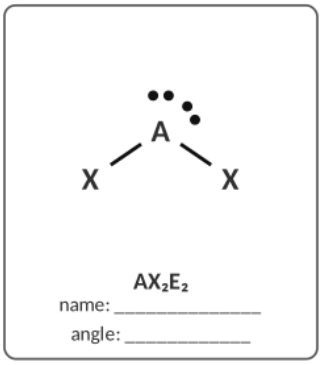
AX_2E
name: _____
angle: _____



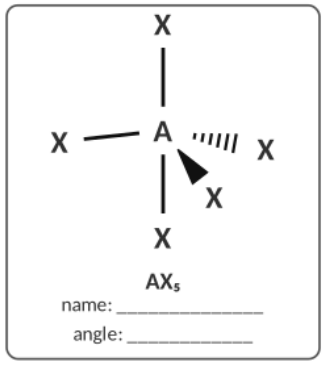
AX_4
name: _____
angle: _____



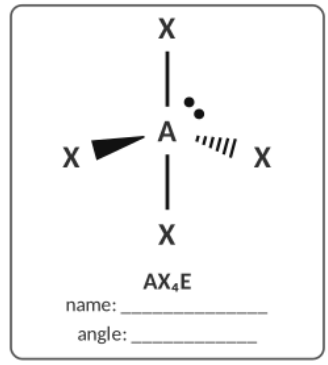
AX_3E
name: _____
angle: _____



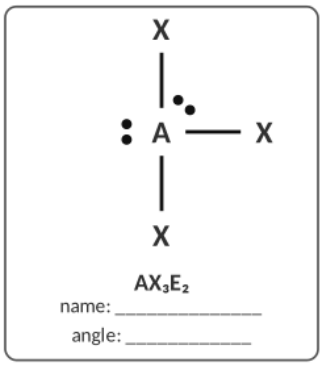
AX_2E_2
name: _____
angle: _____



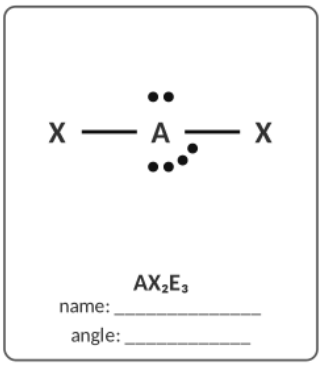
AX_5
name: _____
angle: _____



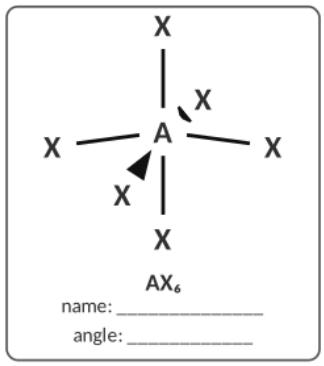
AX_4E
name: _____
angle: _____



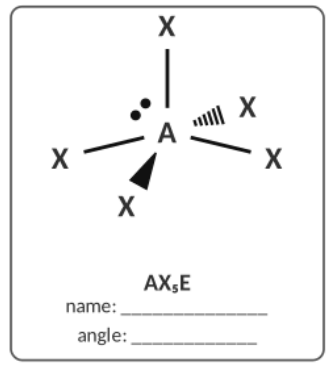
AX_3E_2
name: _____
angle: _____



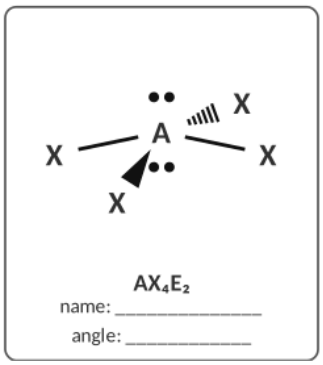
AX_2E_3
name: _____
angle: _____



AX_6
name: _____
angle: _____



AX_5E
name: _____
angle: _____



AX_4E_2
name: _____
angle: _____

A = central atom, X = bonded atoms, E = lone pairs. Write the shape name and the ideal angle in each box.

Name: _____ Date: _____ Period: _____

Total domains	Electron geometry	Lone pairs (E)	Molecular shape
2		0	
3		0 / 1	
4		0 / 1 / 2	
5		0 / 1 / 2 / 3	
6		0 / 1 / 2	

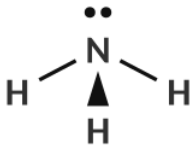
Worked examples

WORKED EXAMPLE — SHAPE OF NH_3

Give the Lewis structure, domains, shape and angle for NH_3.

Work it out here:

Answer: _____



WORKED EXAMPLE — SHAPE OF CO_2

Give the shape and angle of CO_2.

Work it out here:

Answer: _____

Name: _____ Date: _____ Period: _____

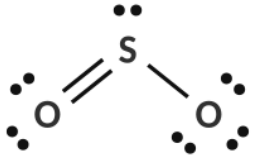
Practice: Lewis structure, shape and angle

Draw each Lewis structure, then name the shape and the bond angle(s).

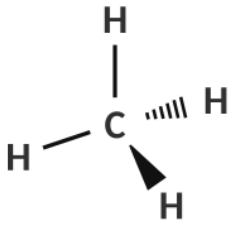
Compound	Lewis structure	Shape	Angle(s)
NO_3^-			
OF_2			
PF_5			
CO_2			
PH_3			
SCl_6			

Name: _____ Date: _____ Period: _____

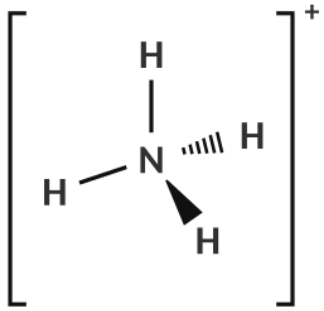
Identify the shape and angle of each structure shown



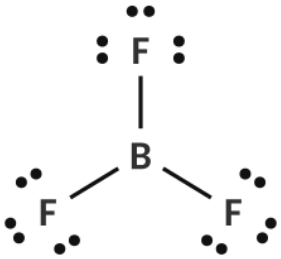
(a)



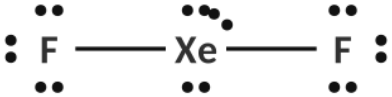
(b)



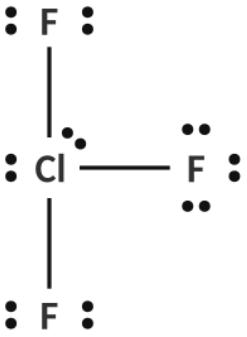
(c)



(d)



(e)



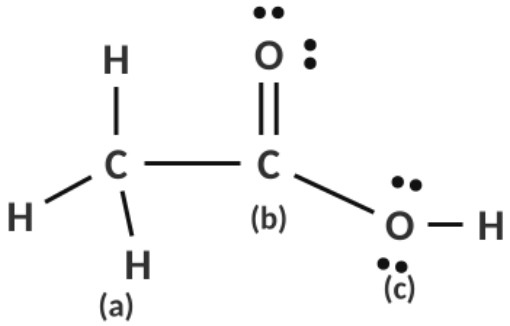
(f)

	Domains on central atom	Electron geometry	Molecular shape	Angle
(a) SO_2				
(b) CH_4				
(c) NH_4^+				
(d) BF_3				
(e) XeF_2				
(f) ClF_3				

Name: _____ Date: _____ Period: _____

Shape around each atom of a larger molecule

Acetic acid (vinegar) has several "central" atoms. Name the shape and bond angle around each atom marked (a), (b) and (c).



[IMAGE GOES HERE]

Optional: the large molecule from the homework

Atom	Domains	Shape	Angle
(a) the CH_3 carbon			
(b) the C=O carbon			
(c) the O–H oxygen			

1. Why is the angle in H_2O (104.5°) smaller than the ideal tetrahedral angle of 109.5°?

2. Both CO_2 and SO_2 have three atoms, but one is linear and the other is bent. Explain the difference.

3. What is the shape of a molecule with 4 electron domains, 2 of which are lone pairs?

A) linear
B) bent
C) trigonal pyramidal
D) tetrahedral

4. PF_5 has what molecular shape?

A) octahedral
B) trigonal planar
C) trigonal bipyramidal
D) square planar

5. What is the approximate bond angle in CH_4?

A) 90°
B) 104.5°
C) 109.5°
D) 120°

6. Which species is trigonal planar?

A) NH_3
B) BF_3
C) PCl_3
D) H_2O

Name: _____ Date: _____ Period: _____

TOPIC
3.5

Bond Polarity & Molecular Polarity

Electronegativity • dipoles • shape decides polarity

BIG IDEA

Electrons in a covalent bond are not always shared equally. The atom with the larger **electronegativity** pulls harder and becomes slightly negative. Whether the **whole molecule** is polar then depends on its **shape**: do the pulls add up or cancel?

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Electronegativity	*Which element has the highest value?*	
Polar covalent bond	*What is unequal about it?*	
Non-polar covalent bond	*When does it happen?*	
Dipole / partial charge (δ+, δ−)	*What does the arrow show?*	
Polar molecule	*Two things that determine it?*	

THINK OF IT LIKE… TUG-OF-WAR

Two teams pull on a rope (the shared electrons). If the teams are equally strong the flag stays in the middle (non-polar). If one team is stronger, the flag shifts toward them (polar). In a molecule with several teams pulling in different directions, it can still end in a draw: if three equal teams pull at 120° from each other, the flag doesn't move (BF_3).

Name: _____ Date: _____ Period: _____

Step 1: the bond

RULE OF THUMB (ΔEN = DIFFERENCE IN ELECTRONEGATIVITY)

ΔEN < 0.4 → **non-polar covalent**

0.4 ≤ ΔEN ≤ 1.7 → **polar covalent**

ΔEN > 1.7 → **ionic** (electron is effectively transferred)

Carbon–hydrogen bonds (ΔEN = 0.35) are treated as non-polar: **do not draw an arrow for C–H**.

Order of the top four: **F > O > Cl > N** (then Br, I, S, C, H).

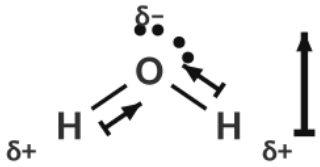
Bond	EN values	ΔEN	Bond type	More electronegative atom
H–H	2.20, 2.20			
C–H	2.55, 2.20			
C–Cl	2.55, 3.16			
N–H	3.04, 2.20			
O–H	3.44, 2.20			
C–O	2.55, 3.44			
Cl–F	3.16, 3.98			
Mg–O	1.31, 3.44			
Na–Cl	0.93, 3.16			

1. Draw a dipole arrow and label δ+ and δ− on each of these bonds: H–Cl, H–O, H–N, C–F, and explain why the C–H bond gets no arrow.

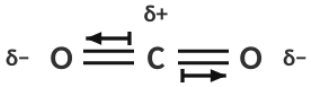
Step 2: the molecule

IS THE MOLECULE POLAR?

1. Draw the Lewis structure and decide on the shape (3.4).
2. Draw a dipole arrow on every polar bond.
3. Ask: **do the arrows cancel?** Are the polar bonds arranged symmetrically (equal arrows pointing in opposite directions)?
4. If they cancel → **non-polar**. If there is a net pull to one side (or any lone pair on the central atom or different atoms attached) → **polar**: label partial charges on every atom.



Water: bent, so the O–H arrows add up: polar.



NON-POLAR
CO_2

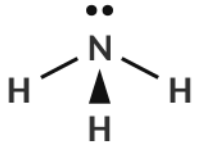
CO_2: two equal arrows in opposite directions cancel: non-polar.

2. CO_2 has polar bonds but is a non-polar molecule, while H_2O has polar bonds and is polar. Explain the difference using shape.

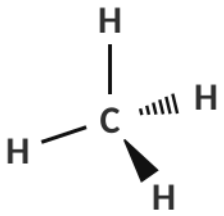
Name: _____ Date: _____ Period: _____

Practice: draw the arrows, decide polar or non-polar

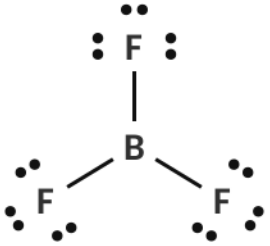
For each molecule, draw an arrow on every polar bond (toward the more electronegative atom), then write POLAR or NON-POLAR and label any partial charges. Use the Lewis structures given.



NH_3

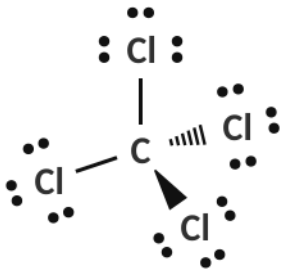


CH_4

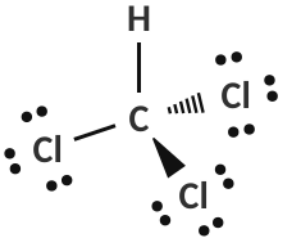


BF_3

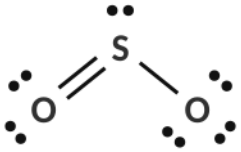
Molecule	Shape	Arrows cancel?	Polar or non-polar?
NH_3			
CH_4			
BF_3			



CCl_4



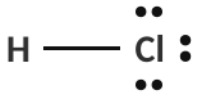
$CHCl_3$



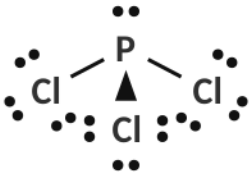
SO_2

Molecule	Shape	Arrows cancel?	Polar or non-polar?
CCl_4			
$CHCl_3$			
SO_2			

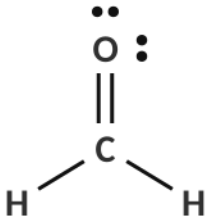
Name: _____ Date: _____ Period: _____



HCl



PCl_3



CH_2O

Molecule	Shape	Arrows cancel?	Polar or non-polar?
HCl			
PCl_3			
CH_2O			

3. Compare CCl_4 and $CHCl_3$. Both are tetrahedral with polar C–Cl bonds. Why is only one polar?

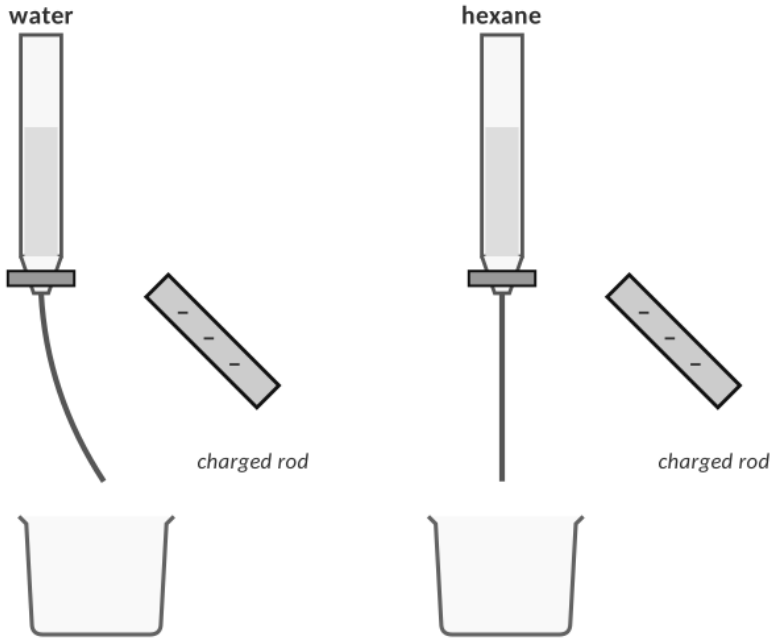
Rules of thumb for shapes

Shape	Same atoms on every position, no lone pairs on the central atom	Otherwise (lone pairs or different atoms)
Linear		
Trigonal planar		
Tetrahedral		
Bent / pyramidal / seesaw / T-shaped		
Octahedral / trigonal bipyramidal		

Name: _____ Date: _____ Period: _____

LAB CORNER • Bending a stream of liquid

A charged plastic rod is held near a thin stream of liquid that is falling from a buret. Water's stream bends toward the rod; hexane's stream does not.



Stream deflection demonstration.

4. Explain the observation for water at the particle level. Draw a quick sketch of how the water molecules line up near the negative rod.

5. Hexane (C_6H_{14}) is not deflected. What does that tell you about hexane?

6. If the rod were positively charged instead, would the water stream bend toward it or away from it? Explain.

7. Which bond is the most polar?

A) C–H
B) H–Cl
C) H–O
D) H–F

8. Which molecule is non-polar even though it has polar bonds?

A) H_2O
B) NH_3
C) CO_2
D) HCl

9. Which molecule is polar?

A) BF_3
B) CCl_4
C) PCl_3
D) CH_4

10. What does the arrow in a bond dipole point toward?

A) the less electronegative atom
B) the more electronegative atom
C) the central atom
D) the nucleus

Name: _____ Date: _____ Period: _____

PRACTICE

Extra Practice — Unit 3

Show all work.

A. Ions and compounds

Element	Valence e^-	Loses or gains?	Ion symbol and charge
Magnesium			
Chlorine			
Oxygen			
Lithium			
Phosphorus			
Calcium			

Metal + non-metal	Ions formed	Formula
potassium + sulfur		
aluminum + chlorine		
magnesium + nitrogen		
calcium + fluorine		
aluminum + oxygen		
sodium + phosphorus		

Name: _____ Date: _____ Period: _____

B. Classify and predict properties

Compound	Ionic or covalent?	Expected state at 25 °C	Conducts when dissolved?
$CaBr_2$			
CH_3OH			
N_2O			
Li_2S			
$C_{12}H_{22}O_{11}$			
$SrCl_2$			

A student tests four unknown white solids. Use the data to decide whether each is ionic or covalent.

Solid	Melting point	Conducts as a solid	Conducts when dissolved	Ionic or covalent?
W	801 °C	no	yes	
X	186 °C	no	no	
Y	2072 °C	no	yes	
Z	54 °C	no	no	

1. Explain how you decided for solid Y, using particles and attractions.

Name: _____ Date: _____ Period: _____

C. Lewis structures, shapes and polarity

Formula	Lewis structure	Shape	Angle	Polar?
H_2S				
CCl_4				
$BeCl_2$				
NF_3				
CS_2				
SF_6				
CH_2Cl_2				
SF_4				

Name: _____ Date: _____ Period: _____

Bond summary: from formula to polarity

Formula	Bond type (ionic / polar covalent / non-polar)	Shape	Polar molecule?
NaCl			
HF			
Cl_2			
PCl_3			
CO_2			
C_2H_6			

D. Explain it

2. Explain why solid NaCl does not conduct electricity but molten NaCl does.

3. A student claims, "Covalent bonds are weaker than ionic bonds, so covalent compounds melt at low temperatures." Evaluate the claim.

4. Draw the Lewis structures of CO_2 and SO_2. Why does one have a bent shape and the other is linear? Which is polar?

5. The bond angle in NH_3 is 107° but in CH_4 it is 109.5°. Explain.

6. Use polarity to explain why oil and water do not mix. (Hint: think about the stream demo.)

E. Multiple choice

7. Which compound contains BOTH ionic and covalent bonds?

A) NaCl
B) NH_4NO_3
C) CO_2
D) CH_4

8. What is the shape of the sulfate ion, SO_4^{2-}?

A) trigonal planar
B) tetrahedral
C) trigonal pyramidal
D) octahedral

9. Which of the following molecules has a triple bond?

A) O_2
B) N_2
C) Cl_2
D) H_2O

10. A molecule with the shape ‘seesaw’ has how many electron domains on the central atom?

A) 3
B) 4
C) 5
D) 6

11. Which pair has the greater ΔEN?

A) C–H
B) N–H
C) O–H
D) Cl–Cl

Name: _____ Date: _____ Period: _____

TOPIC

★

Unit 3 Review

Put it in your own words before the test

1. Compare ionic and covalent bonding: which elements, what happens to the electrons, and what the compounds are like.

2. Describe the five steps you follow to go from a formula to a Lewis structure.

3. What does VSEPR stand for and how do you use it to predict a shape?

4. Explain how a molecule can have polar bonds but be non-polar. Give an example.

Name: _____ Date: _____ Period: _____

Concept map

Make your own concept map using: octet, ionic, covalent, Lewis structure, VSEPR, shape, polarity, melting point